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1. Introduction

Customer churn—the act of clients discontinuing their relationship with a financial service provider—is a critical issue in the credit card market. Retaining existing customers is significantly more cost-effective than acquiring new ones, and churn can lead to substantial financial losses, diminished brand equity, and a reduced customer lifetime value (Kotler & Keller, 2016). In an increasingly saturated banking landscape, understanding why customers leave has become a top priority. Predictive modelling offers an opportunity to uncover patterns in churn behaviour, enabling institutions to proactively address customer dissatisfaction and strengthen retention strategies (Verbeke et al., 2011). This proposal explores the application of predictive modelling techniques to identify the key drivers of churn in the credit card sector, ultimately empowering banks to make data-driven decisions to mitigate revenue loss and improve customer loyalty.

2. Problem Statement

Churn in the credit card market is an ongoing and costly issue that continues to impact banks' operational efficiency and profitability. Research by Neslin et al. (2006) suggests that customer attrition rates in financial services can range from 15% to 25% annually, resulting in considerable revenue loss and increased acquisition costs. Traditional retention strategies often fail to detect at-risk customers before they leave, leading to reactive rather than proactive interventions. As competition intensifies and customer expectations evolve, banks must leverage data analytics to better understand churn patterns and customer behaviours. This project focuses on investigating the churn phenomenon using a real-world dataset of banking customer profiles to identify which behavioural and demographic factors are most associated with customer departure. The ultimate goal is to enable financial institutions to anticipate churn with greater accuracy and implement timely, cost-effective retention strategies.

3. Project Objective

The objective of this project is to develop a predictive model that identifies customers at high risk of churning in the credit card market, addressing a critical business need for improved customer retention. This objective is **Specific**, as it focuses on predicting customer churn using demographic, behavioural, and transactional features from a real-world banking dataset. It is **Measurable** through the use of standard evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to assess the model's performance. The objective is **Achievable** by applying well-established machine learning techniques—such as logistic regression, decision trees, and ensemble methods—implemented using Python libraries like scikit-learn and pandas, within the technical scope and resources available. It is **Relevant** to the financial industry's growing need to reduce attrition, increase profitability, and enhance customer loyalty. Finally, the objective is **Time-bound**, with the model development, validation, and reporting to be completed within a 12-week timeframe, culminating in submission by June 2025. Prior to model development, exploratory data analysis (EDA) will be conducted to examine feature distributions, identify outliers, and address missing or imbalanced data. This step ensures clean input and reliable model outputs.

4. Data Identification

- **Dataset Name:** Credit Card Customer
- **Source/Link:** [<https://www.kaggle.com/datasets/sakshigoyal7/credit-card-customers/data>]
- **Description:** The dataset comprises 10,127 records and 23 columns detailing credit card customer information, including demographics, account status, product usage, and transaction behaviour. It contains a mix of **categorical** and **numerical** variables across various measurement scales. **Nominal variables**, such as *Gender*, and *Marital_Status*, represent labelled categories without inherent order. **Ordinal variables**, like *Education_Level* and *Income_Category*, have a ranked structure (e.g., Uneducated < Graduate < Doctorate). **Interval variables**, such as *Customer_Age*, reflect numerical data with equal intervals but no true zero. **Ratio variables**, including *Credit_Limit*, *Total_Trans_Amt*, and *Avg_Utilization_Ratio*, possess a meaningful zero and support full arithmetic operations. The target variable, *Attrition_Flag*, is binary, indicating whether a customer has churned or remains active. The column *CLIENTNUM* is a customer ID and not used in model training. The final two columns, generated by a pre-trained Naive Bayes classifier, are excluded based on the

dataset author's note. After these exclusions, **19 predictive features** remain suitable for model development.

- **Justification for Selection:** This dataset was selected for its strong alignment with predictive modelling tasks in a financial context. It includes a rich variety of customer-level features commonly associated with behavioural analytics, allowing for exploration of meaningful patterns. The real-world nature of the data, coupled with its clean structure and sufficient sample size, makes it ideal for training, validating, and interpreting classification models. Its relevance to customer engagement, credit management, and transaction behaviour enables the development of robust, business-focused predictive solutions.
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5. Target Variable Selection

- **Target Variable Name:** *Attrition Flag*
 - **Description and Contextual Relevance:** The *Attrition_Flag* variable indicates whether a customer has churned (labelled as "Attrited Customer") or remains with the credit card provider ("Existing Customer"). It is a binary categorical variable and serves as the dependent variable in this project. This variable is central to the business problem, as customer attrition directly impacts profitability, customer lifetime value, and operational planning within the financial sector. Accurate prediction of churn enables institutions to take proactive steps—such as targeted retention campaigns, personalised incentives, or credit adjustments—before customer departure occurs. Misclassification could lead to inefficient use of marketing resources or missed opportunities to retain valuable customers. On the other hand, reliable churn predictions can guide strategic decisions and improve overall customer experience, contributing to long-term sustainability and competitive advantage for credit card providers.
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6. Prediction Type

- **Prediction Type:** Decision
- **Justification:** This project adopts a decision-type prediction, as the target variable (*Attrition_Flag*) is categorical and binary—indicating whether a customer has churned or remained active. The goal is to make a clear classification decision for each customer: churn or not churn. This aligns with

the business objective of identifying at-risk customers to inform retention strategies. Unlike estimation (which predicts continuous values) or ranking (which prioritises outcomes), decision-type prediction delivers discrete outcomes that directly support actionable interventions, such as targeted offers or personalised follow-ups. This makes it the most suitable and practical prediction type for this context.

7. Potential Applications

The predictive model developed in this project has broad real-world applications, particularly for banks and credit card companies seeking to address customer churn. One of its primary uses is in **targeted retention strategies**—by accurately identifying customers who are at high risk of attrition, financial institutions can take timely actions such as offering customised incentives, adjusting credit terms, or delivering personalised communications to retain valuable clients. These proactive interventions help reduce churn-related revenue losses and improve customer lifetime value.

Furthermore, insights from the model can support **customer segmentation**, allowing marketing teams to tailor campaigns based on risk levels and behavioural profiles. High-value customers identified as likely to churn can be prioritised for retention, while low-risk customers can be nurtured through loyalty-building initiatives.

Importantly, the model can be integrated into **Customer Relationship Management (CRM) systems**, such as Salesforce or HubSpot. CRM platforms are used to manage customer interactions and store relationship data. By embedding predictive outputs into these systems, banks can automate alerts, track churn risk in real time, and guide frontline staff in delivering more relevant and timely engagement.

Overall, the model enables **data-driven decision-making**, enhances operational efficiency, and empowers institutions to foster long-term customer relationships in a highly competitive financial landscape.

8. References

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